KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2016
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the terms: cryptography, cryptanalysis. Discuss Shannon characteristics of good cipher. (10)
   b) Define known-plaintext attack. How can it be applied for RSA cryptosystem? (10)
   c) Explain with example: a) substitution ciphers, b) transposition ciphers. Discuss a specific key exchange protocol to distribute a key. (10)
   d) How is trade secret protection vanished by 'reverse engineering'? Explain. (05)

2. a) What are the characteristics of digital signature? Explain ElGamal digital signature algorithm with an example. (12)
   b) Define cryptographic hash function. Discuss some properties of it. (08)
   c) Explain how $ed = 1$ in RSA encryption algorithm? [Where d and e have their usual meanings]. (07)
   d) Digital signature and public key cryptography can be combined. Why and how? Explain. (08)

3. a) Discuss the steps to sign documents with public-key cryptography and one way hash functions. (07)
   b) What are the parameters that make a computer network vulnerable? Discuss them briefly. (10)
   c) What are the limitations of RSA cryptosystem? Make a comparison between RSA and ElGamal cryptosystems. (10)
   d) Explain the terms: (i) denial-of-service (DOS) attack and (ii) distributed denial-of-service (DDOS) attack. (08)

4. a) Discuss the threats to E-mail. Also discuss how to defend against it. (07)
   b) Define sensitive data. Discuss the factors that can make data sensitive. (10)
   c) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)
   d) What is meant by homomorphic encryption? Explain with examples. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) If two users share access to a segment, they must do so by the same name. Must their protection rights to it be the same? Why or why not? (05)
   b) Why is Kerberos a credible security solution? Describe, how is authentication and access authorization implemented in Kerberos. (10)
   c) Make short notes over (i) Loose-lipped systems (ii) Access control matrix (08)
   d) Briefly explain the operational modes of Intrusion detection system. (12)

6. a) What do you mean by Wired Equivalent Privacy (WEP)? Briefly explain all the possible attacks on wifi protected access (WPA & WPA2) networks. (09)
   b) Draw and explain each of the constructing components of wireless data frame. (09)
   c) Suppose Alice is sending an Email to Bob, but Alice wants her Email to be anonymous. How can Alice achieve this? What are the threats about which Alice is afraid of? (12)
   d) Describe how RSA SecurID suite works. (05)

7. a) Suppose, while browsing web, your machine got attacked by a virus. What are the best places for that virus to make itself hidden and gain control over your system? Describe the situation before and after this infection. (09)
   b) What are the error detecting codes that can be used to detect when digital objects are surreptitiously altered? (10)

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c) Suppose, an attacker wants to attack a target server which has higher bandwidth than the attacker. Explain an attack with necessary figure by which the attacker can overflow the target server's maximum connection capacity. (10)

d) Write down a short note on Smurf Attack. (06)

8. a) What do you mean by security through obscurity and challenge-response system? Briefly explain. (10)

b) How would you distinguish a risk from other project events? Discuss the strategies for dealing with risks. (10)

c) What type of attacks "Keystroke loggers" is? Briefly explain. (08)

d) Suppose, a system is being accessed by different users with different privileges. What are the principles that you as an administrator would adopt while giving access to the users. (07)

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